

Introduction to WordNet: An On-line Lexical Database

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(Revised August 1993)

WordNet is an on-line lexical reference system whose design is inspired by current psycholinguistic theories of human lexical memory. English nouns, verbs, and adjectives are organized into synonym sets, each representing one underlying lexical concept. Different relations link the synonym sets.

Standard alphabetical procedures for organizing lexical information put together words that are spelled alike and scatter words with similar or related meanings haphazardly through the list. Unfortunately, there is no obvious alternative, no other simple way for lexicographers to keep track of what has been done or for readers to find the word they are looking for. But a frequent objection to this solution is that finding things on an alphabetical list can be tedious and time-consuming. Many people who would like to refer to a dictionary decide not to bother with it because finding the information would interrupt their work and break their train of thought.

In this age of computers, however, there is an answer to that complaint. One obvious reason to resort to on-line dictionaries—lexical databases that can be read by computers—is that computers can search such alphabetical lists much faster than people can. A dictionary entry can be available as soon as the target word is selected or typed into the keyboard. Moreover, since dictionaries are printed from tapes that are read by computers, it is a relatively simple matter to convert those tapes into the appropriate kind of lexical database. Putting conventional dictionaries on line seems a simple and natural marriage of the old and the new.

Once computers are enlisted in the service of dictionary users, however, it quickly becomes apparent that it is grossly inefficient to use these powerful machines as little more than rapid page-turners. The challenge is to think what further use to make of them. WordNet is a proposal for a more effective combination of traditional lexicographic information and modern high-speed computation.

This, and the accompanying four papers, is a detailed report of the state of WordNet as of 1990. In order to reduce unnecessary repetition, the papers are written to be read consecutively.

Psycholexicology

Murray's *Oxford English Dictionary* (1928) was compiled “on historical principles” and no one doubts the value of the *OED* in settling issues of word use or sense priority. By focusing on historical (diachronic) evidence, however, the *OED*, like other standard dictionaries, neglected questions concerning the synchronic organization of lexical knowledge.

It is now possible to envision ways in which that omission might be repaired. The 20th Century has seen the emergence of psycholinguistics, an interdisciplinary field of research concerned with the cognitive bases of linguistic competence. Both linguists and psycholinguists have explored in considerable depth the factors determining the contemporary (synchronic) structure of linguistic knowledge in general, and lexical knowledge in particular—Miller and Johnson-Laird (1976) have proposed that research concerned with the lexical component of language should be called psycholexicology. As linguistic theories evolved in recent decades, linguists became increasingly explicit about the information a lexicon must contain in order for the phonological, syntactic, and lexical components to work together in the everyday production and comprehension of linguistic messages, and those proposals have been incorporated into the work of psycholinguists. Beginning with word association studies at the turn of the century and continuing down to the sophisticated experimental tasks of the past twenty years, psycholinguists have discovered many synchronic properties of the mental lexicon that can be exploited in lexicography.

In 1985 a group of psychologists and linguists at Princeton University undertook to develop a lexical database along lines suggested by these investigations (Miller, 1985). The initial idea was to provide an aid to use in searching dictionaries conceptually, rather than merely alphabetically—it was to be used in close conjunction with an on-line dictionary of the conventional type. As the work proceeded, however, it demanded a more ambitious formulation of its own principles and goals. WordNet is the result. Inasmuch as it instantiates hypotheses based on results of psycholinguistic research, WordNet can be said to be a dictionary based on psycholinguistic principles.

How the leading psycholinguistic theories should be exploited for this project was not always obvious. Unfortunately, most research of interest for psycholexicology has dealt with relatively small samples of the English lexicon, often concentrating on nouns at the expense of other parts of speech. All too often, an interesting hypothesis is put forward, fifty or a hundred words illustrating it are considered, and extension to the rest of the lexicon is left as an exercise for the reader. One motive for developing WordNet was to expose such hypotheses to the full range of the common vocabulary. WordNet presently contains approximately 95,600 different word forms (51,500 simple words and 44,100 collocations) organized into some 70,100 word meanings, or sets of synonyms, and only the most robust hypotheses have survived.

The most obvious difference between WordNet and a standard dictionary is that WordNet divides the lexicon into five categories: nouns, verbs, adjectives, adverbs, and function words. Actually, WordNet contains only nouns, verbs, adjectives, and adverbs.¹ The relatively small set of English function words is omitted on the assumption (supported by observations of the speech of aphasic patients: Garrett, 1982) that they are probably stored separately as part of the syntactic component of language. The realization that syntactic categories differ in subjective organization emerged first from studies of word associations. Fillenbaum and Jones (1965), for example, asked English-

¹ A discussion of adverbs is not included in the present collection of papers.

speaking subjects to give the first word they thought of in response to highly familiar words drawn from different syntactic categories. The modal response category was the same as the category of the probe word: noun probes elicited nouns responses 79% of the time, adjectives elicited adjectives 65% of the time, and verbs elicited verbs 43% of the time. Since grammatical speech requires a speaker to know (at least implicitly) the syntactic privileges of different words, it is not surprising that such information would be readily available. How it is learned, however, is more of a puzzle: it is rare in connected discourse for adjacent words to be from the same syntactic category, so Fillenbaum and Jones's data cannot be explained as association by contiguity.

The price of imposing this syntactic categorization on WordNet is a certain amount of redundancy that conventional dictionaries avoid—words like *back*, for example, turn up in more than one category. But the advantage is that fundamental differences in the semantic organization of these syntactic categories can be clearly seen and systematically exploited. As will become clear from the papers following this one, nouns are organized in lexical memory as topical hierarchies, verbs are organized by a variety of entailment relations, and adjectives and adverbs are organized as N-dimensional hyperspaces. Each of these lexical structures reflects a different way of categorizing experience; attempts to impose a single organizing principle on all syntactic categories would badly misrepresent the psychological complexity of lexical knowledge.

The most ambitious feature of WordNet, however, is its attempt to organize lexical information in terms of word meanings, rather than word forms. In that respect, WordNet resembles a thesaurus more than a dictionary, and, in fact, Laurence Urdang's revision of Rodale's *The Synonym Finder* (1978) and Robert L. Chapman's revision of *Roget's International Thesaurus* (1977) have been helpful tools in putting WordNet together. But neither of those excellent works is well suited to the printed form. The problem with an alphabetical thesaurus is redundant entries: if word W_x and word W_y are synonyms, the pair should be entered twice, once alphabetized under W_x and again alphabetized under W_y . The problem with a topical thesaurus is that two look-ups are required, first on an alphabetical list and again in the thesaurus proper, thus doubling a user's search time. These are, of course, precisely the kinds of mechanical chores that a computer can perform rapidly and efficiently.

WordNet is not merely an on-line thesaurus, however. In order to appreciate what more has been attempted in WordNet, it is necessary to understand its basic design (Miller and Fellbaum, 1991).

The Lexical Matrix

Lexical semantics begins with a recognition that a word is a conventional association between a lexicalized concept and an utterance that plays a syntactic role. This definition of "word" raises at least three classes of problems for research. First, what kinds of utterances enter into these lexical associations? Second, what is the nature and organization of the lexicalized concepts that words can express? Third, what syntactic roles do different words play? Although it is impossible to ignore any of these questions while considering only one, the emphasis here will be on the second class of

problems, those dealing with the semantic structure of the English lexicon.

Since the word “word” is commonly used to refer both to the utterance and to its associated concept, discussions of this lexical association are vulnerable to terminological confusion. In order to reduce ambiguity, therefore, “word form” will be used here to refer to the physical utterance or inscription and “word meaning” to refer to the lexicalized concept that a form can be used to express. Then the starting point for lexical semantics can be said to be the mapping between forms and meanings (Miller, 1986). A conservative initial assumption is that different syntactic categories of words may have different kinds of mappings.

Table 1 is offered simply to make the notion of a lexical matrix concrete. Word forms are imagined to be listed as headings for the columns; word meanings as headings for the rows. An entry in a cell of the matrix implies that the form in that column can be used (in an appropriate context) to express the meaning in that row. Thus, entry $E_{1,1}$ implies that word form F_1 can be used to express word meaning M_1 . If there are two entries in the same column, the word form is polysemous; if there are two entries in the same row, the two word forms are synonyms (relative to a context).

Table 1

Illustrating the Concept of a Lexical Matrix:

F_1 and F_2 are synonyms; F_2 is polysemous

Word Meanings	Word Forms				
	F_1	F_2	F_3	. . .	F_n
M_1	$E_{1,1}$	$E_{1,2}$			
M_2		$E_{2,2}$			
M_3			$E_{3,3}$		
$\vdots$				$\ddots$	
M_m					$E_{m,n}$

Mappings between forms and meanings are many:many—some forms have several different meanings, and some meanings can be expressed by several different forms. Two difficult problems of lexicography, polysemy and synonymy, can be viewed as complementary aspects of this mapping. That is to say, polysemy and synonymy are problems that arise in the course of gaining access to information in the mental lexicon: a listener or reader who recognizes a form must cope with its polysemy; a speaker or writer who hopes to express a meaning must decide between synonyms.

As a parenthetical comment, it should be noted that psycholinguists frequently represent their hypotheses about language processing by box-and-arrow diagrams. In that notation, a lexical matrix could be represented by two boxes with arrows going between them in both directions. One box would be labeled ‘Word Meaning’ and the other ‘Word Form’; arrows would indicate that a language user could start with a meaning and look for appropriate forms to express it, or could start with a form and

retrieve appropriate meanings. This box-and-arrow representation makes clear the difference between meaning:meaning relations (in the Word Meaning box) and word:word relations (in the Word Form box). In its initial conception, WordNet was concerned solely with the pattern of semantic relations between lexicalized concepts; that is to say, it was to be a theory of the Word Meaning box. As work proceeded, however, it became increasingly clear that lexical relations in the Word Form box could not be ignored. At present, WordNet distinguishes between semantic relations and lexical relations; the emphasis is still on semantic relations between meanings, but relations between words are also included.

Although the box-and-arrow representation respects the difference between these two kinds of relations, it has the disadvantage that the intricate details of the many:many mapping between meanings and forms are slighted, which not only conceals the reciprocity of polysemy and synonymy, but also obscures the major device used in WordNet to represent meanings. For that reason, this description of WordNet has been introduced in terms of a lexical matrix, rather than as a box-and-arrow diagram.

How are word meanings represented in WordNet? In order to simulate a lexical matrix it is necessary to have some way to represent both forms and meanings in a computer. Inscriptions can provide a reasonably satisfactory solution for the forms, but how meanings should be represented poses a critical question for any theory of lexical semantics. Lacking an adequate psychological theory, methods developed by lexicographers can provide an interim solution: definitions can play the same role in a simulation that meanings play in the mind of a language user.

How lexicalized concepts are to be represented by definitions in a theory of lexical semantics depends on whether the theory is intended to be constructive or merely differential. In a constructive theory, the representation should contain sufficient information to support an accurate construction of the concept (by either a person or a machine). The requirements of a constructive theory are not easily met, and there is some reason to believe that the definitions found in most standard dictionaries do not meet them (Gross, Kegl, Gildea, and Miller, 1989; Miller and Gildea, 1987). In a differential theory, on the other hand, meanings can be represented by any symbols that enable a theorist to distinguish among them. The requirements for a differential theory are more modest, yet suffice for the construction of the desired mappings. If the person who reads the definition has already acquired the concept and needs merely to identify it, then a synonym (or near synonym) is often sufficient. In other words, the word meaning M_1 in Table 1 can be represented by simply listing the word forms that can be used to express it: $\{F_1, F_2, \dots\}$. (Here and later, the curly brackets, '{' and '}', surround the sets of synonyms that serve as identifying definitions of lexicalized concepts.) For example, someone who knows that *board* can signify either a piece of lumber or a group of people assembled for some purpose will be able to pick out the intended sense with no more help than *plank* or *committee*. The synonym sets, $\{board, plank\}$ and $\{board, committee\}$ can serve as unambiguous designators of these two meanings of *board*. These synonym sets (synsets) do not explain what the concepts are; they merely signify that the concepts exist. People who know English are assumed to have already acquired

the concepts, and are expected to recognize them from the words listed in the synset.

A lexical matrix, therefore, can be represented for theoretical purposes by a mapping between written words and synsets. Since English is rich in synonyms, synsets are often sufficient for differential purposes. Sometimes, however, an appropriate synonym is not available, in which case the polysemy can be resolved by a short gloss, e.g., {*board*, (a person's meals, provided regularly for money)} can serve to differentiate this sense of *board* from the others; it can be regarded as a synset with a single member. The gloss is not intended for use in constructing a new lexical concept by someone not already familiar with it, and it differs from a synonym in that it is not used to gain access to information stored in the mental lexicon. It fulfills its purpose if it enables the user of WordNet, who is assumed to know English, to differentiate this sense from others with which it could be confused.

Synonymy is, of course, a lexical relation between word forms, but because it is assigned this central role in WordNet, a notational distinction is made between words related by synonymy, which are enclosed in curly brackets, '{' and '}', and other lexical relations, which will be enclosed in square brackets, '[' and ']'. Semantic relations are indicated by pointers.

WordNet is organized by semantic relations. Since a semantic relation is a relation between meanings, and since meanings can be represented by synsets, it is natural to think of semantic relations as pointers between synsets. It is characteristic of semantic relations that they are reciprocated: if there is a semantic relation *R* between meaning {*x*, *x'*, . . .} and meaning {*y*, *y'*, . . .}, then there is also a relation *R'* between {*y*, *y'*, . . .} and {*x*, *x'*, . . .}. For the purposes of the present discussion, the names of the semantic relations will serve a dual role: if the relation between the meanings {*x*, *x'*, . . .} and {*y*, *y'*, . . .} is called *R*, then *R* will also be used to designate the relation between individual word forms belonging to those synsets. It might be logically tidier to introduce separate terms for the relation between meanings and for the relation between forms, but even greater confusion might result from the introduction of so many new technical terms.

The following examples illustrate (but do not exhaust) the kinds of relations used to create WordNet.

Synonymy

From what has already been said, it should be obvious that the most important relation for WordNet is similarity of meaning, since the ability to judge that relation between word forms is a prerequisite for the representation of meanings in a lexical matrix. According to one definition (usually attributed to Leibniz) two expressions are synonymous if the substitution of one for the other never changes the truth value of a sentence in which the substitution is made. By that definition, true synonyms are rare, if they exist at all. A weakened version of this definition would make synonymy relative to a context: two expressions are synonymous in a linguistic context *C* if the substitution of one for the other in *C* does not alter the truth value. For example, the substitution of *plank* for *board* will seldom alter truth values in carpentry contexts, although there are other contexts of *board* where that substitution would be totally inappropriate.

Note that the definition of synonymy in terms of substitutability makes it necessary to partition WordNet into nouns, verbs, adjectives, and adverbs. That is to say, if concepts are represented by synsets, and if synonyms must be interchangeable, then words in different syntactic categories cannot be synonyms (cannot form synsets) because they are not interchangeable. Nouns express nominal concepts, verbs express verbal concepts, and modifiers provide ways to qualify those concepts. In other words, the use of synsets to represent word meanings is consistent with psycholinguistic evidence that nouns, verbs, and modifiers are organized independently in semantic memory. An argument might be made in favor of still further partitions: some words in the same syntactic category (particularly verbs) express very similar concepts, yet cannot be interchanged without making the sentence ungrammatical.

The definition of synonymy in terms of truth values seems to make synonymy a discrete matter: two words either are synonyms or they are not. But as some philosophers have argued, and most psychologists accept without considering the alternative, synonymy is best thought of as one end of a continuum along which similarity of meaning can be graded. It is probably the case that semantically similar words can be interchanged in more contexts than can semantically dissimilar words. But the important point here is that theories of lexical semantics do not depend on truth-functional conceptions of meaning; semantic similarity is sufficient. It is convenient to assume that the relation is symmetric: if x is similar to y , then y is equally similar to x .

The gradability of semantic similarity is ubiquitous, but it is most important for understanding the organization of adjectival and adverbial meanings.

Antonymy

Another familiar relation is antonymy, which turns out to be surprisingly difficult to define. The antonym of a word x is sometimes *not- x* , but not always. For example, *rich* and *poor* are antonyms, but to say that someone is not rich does not imply that they must be poor; many people consider themselves neither rich nor poor. Antonymy, which seems to be a simple symmetric relation, is actually quite complex, yet speakers of English have little difficulty recognizing antonyms when they see them.

Antonymy is a lexical relation between word forms, not a semantic relation between word meanings. For example, the meanings $\{rise, ascend\}$ and $\{fall, descend\}$ may be conceptual opposites, but they are not antonyms; $[rise/fall]$ are antonyms and so are $[ascend/descend]$, but most people hesitate and look thoughtful when asked if *rise* and *descend*, or *ascend* and *fall*, are antonyms. Such facts make apparent the need to distinguish between semantic relations between word forms and semantic relations between word meanings. Antonymy provides a central organizing principle for the adjectives and adverbs in WordNet, and the complications that arise from the fact that antonymy is a semantic relation between words are better discussed in that context.

Hyponymy

Unlike synonymy and antonymy, which are lexical relations between word forms, hyponymy/hypernymy is a semantic relation between word meanings: e.g., *{maple}* is a hyponym of *{tree}*, and *{tree}* is a hyponym of *{plant}*. Much attention has been devoted to hyponymy/hypernymy (variously called subordination/superordination, subset/superset, or the ISA relation). A concept represented by the synset $\{x, x', \dots\}$ is said to be a hyponym of the concept represented by the synset $\{y, y', \dots\}$ if native speakers of English accept sentences constructed from such frames as *An x is a (kind of) y* . The relation can be represented by including in $\{x, x', \dots\}$ a pointer to its superordinate, and including in $\{y, y', \dots\}$ pointers to its hyponyms.

Hyponymy is transitive and asymmetrical (Lyons, 1977, vol. 1), and, since there is normally a single superordinate, it generates a hierarchical semantic structure, in which a hyponym is said to be below its superordinate. Such hierarchical representations are widely used in the construction of information retrieval systems, where they are called inheritance systems (Touretzky, 1986): a hyponym inherits all the features of the more generic concept and adds at least one feature that distinguishes it from its superordinate and from any other hyponyms of that superordinate. For example, *maple* inherits the features of its superordinate, *tree*, but is distinguished from other trees by the hardness of its wood, the shape of its leaves, the use of its sap for syrup, etc. This convention provides the central organizing principle for the nouns in WordNet.

Meronymy

Synonymy, antonymy, and hyponymy are familiar relations. They apply widely throughout the lexicon and people do not need special training in linguistics in order to appreciate them. Another relation sharing these advantages—a semantic relation—is the part-whole (or HASA) relation, known to lexical semanticists as meronymy/holonymy. A concept represented by the synset $\{x, x', \dots\}$ is a meronym of a concept represented by the synset $\{y, y', \dots\}$ if native speakers of English accept sentences constructed from such frames as *A y has an x (as a part)* or *An x is a part of y* . The meronymic relation is transitive (with qualifications) and asymmetrical (Cruse, 1986), and can be used to construct a part hierarchy (with some reservations, since a meronym can have many holonyms). It will be assumed that the concept of a part of a whole can be a part of a concept of the whole, although it is recognized that the implications of this assumption deserve more discussion than they will receive here.

These and other similar relations serve to organize the mental lexicon. They can be represented in WordNet by parenthetical groupings or by pointers (labeled arcs) from one synset to another. These relations represent associations that form a complex network; knowing where a word is situated in that network is an important part of knowing the word's meaning. It is not profitable to discuss these relations in the abstract, however, because they play different roles in organizing the lexical knowledge associated with different syntactic categories.

Morphological Relations

An important class of lexical relations are the morphological relations between word forms. Initially, interest was limited to semantic relations; no plans were made to include morphological relations in WordNet. As work progressed, however, it became increasingly obvious that if WordNet was to be of any practical use to anyone, it would have to deal with inflectional morphology. For example, if someone put the computer's cursor on the word *trees* and clicked a request for information, WordNet should not reply that the word was not in the database. A program was needed to strip off the plural suffix and then to look up *tree*, which certainly is in the database. This need led to the development of a program for dealing with inflectional morphology.

Although the inflectional morphology of English is relatively simple, writing a computer program to deal with it proved to be a more complex task than had been expected. Verbs are the major problem, of course, since there are four forms and many irregular verbs. But the software has been written and is presently available as part of the interface between the lexical database and the user. In the course of this development it became obvious that programs dealing with derivational morphology would greatly enhance the value of WordNet, but that more ambitious project has not yet been undertaken.

The three papers following this introduction have little to say about lexical relations resulting from inflectional morphology, since those relations are incorporated in the interface to WordNet, not in the central database.